

Assignment #1

Recommended Reading:

- Diggle, Heagerty, Liang and Zeger (DHLZ) Chapter 1
 - Burton, Gurrin and Sly (1998) Tutorial in biostatistics: Extending the simple linear regression model to account for correlated responses. *Statistics in Medicine*, 17, 1261-91: sections 1, 2.1-2.4. (*Posted on the Canvas page in Files → Resources → Miscellaneous*)
- 1) Find a published scientific article of interest to you that exhibits correlated data. Read the article and answer the following questions.
- a. What was the primary scientific question?
 - b. What was the study design? Did it involve random sampling from a population, cluster sampling, no sampling? Were repeated measurements taken on individual subjects? What feature(s) of the study design resulted in correlated data?
 - c. Name the primary outcome variable? What type of outcome is it? (binary, ordinal, ...)
 - d. What is the primary sampling unit (psu)? How many are there? Were all the psu's observed or did some have missing data?
 - e. What were the elementary units (eu) or sub-units? How many are there in total?
 - f. Is the number of units the same in each cluster (i.e., $n_i = n$) without accounting for missing responses? If missing responses are accounted for?
 - g. Did the authors state the magnitude of the within-cluster correlation (ρ)? If yes, what is it? If not, what value would you guess for ρ ? Why? Is it reasonable that the correlation would be positive or negative?
 - h. How did the correlation arise? What mechanism do you think caused the correlation?
 - i. Name one factor of interest to the investigators. (You can choose a treatment factor or a covariate.) Does this factor characterize units with the cluster or does it characterize an entire cluster (i.e., all the objects within a cluster have the same value of the factor)?
 - j. State the population characteristic of primary interest to the investigators (e.g., mean, prevalence, odds ratio, etc.)
 - k. Name the primary analysis reported in the paper. Did this analysis account for the correlation in the outcomes?
 - l. Were there any points regarding the design or analysis that were obscure to you?

- 2) Consider the variance inflation factor: $VIF = 1 + (n-1) \rho$. (See Lecture Notes 2.)
 - a. For what value of ρ is VIF largest? For this value of ρ , what is the value of VIF, and what, therefore, is the variance of $\bar{Y}_i$? What, then is the “effective sample size,” i.e., the sample size for uncorrelated outcomes that would give the same value for the variance?
 - b. For what value of ρ is VIF equal to 1? For this situation, what is the variance of $\bar{Y}_i$? What is the “effective sample size” here? For what value of ρ is VIF less than 1? What is the minimum value of VIF? (Hint: What is the minimum value of the variance of the cluster means?)

Supplementary Exercises (You are encouraged to do/ all of these.)

- 3) Give an example where you might expect the within cluster correlation to be negative. What would be the impact of the correlation on statistical inference in this example?
- 4) With uncorrelated data, doubling the total sample size does what to the variance of the overall average? With correlated data within clusters, we need to be more specific about what we mean by doubling the total sample size. One way would be to double the number of clusters. Another way would be to double the cluster size. (Other ways would involve increasing both.) Assume the correlation parameter, ρ , is positive.
 - a. Doubling the total sample size by doubling the cluster size does what to the overall average?
 - b. Doubling the sample size by doubling the number of clusters does what to the variance of the overall average?
 - c. Using your results, which strategy reduces the variance more: doubling the cluster size or doubling the number of clusters?
 - d. How would you explain this phenomenon to someone who reasons that additional observations should count the same (for reducing variance) “regardless of which cluster they are in?”